



3813ICT - Full Stack Development Assignment

Individual Assignment (70%)

Phase 1 – Specification, Design, Prototype (25%)

Due: 5pm Thursday 27 August 2026 (Week 6)

Phase 2 – Fully functioning application (45%)

Due: 5pm Friday 02 October 2026 (Week 11)

- **Late Submissions:** Late submission is allowed but a penalty applies. The penalty is defined as the reduction of the mark allocated to the assessment item by 5% of the total weighted mark for the assessment item, for each day that the item is late. Assessment items submitted more than seven days after the due date will be awarded zero marks.
- **Extensions:** You can request an extension of time.
 - Check section 3.8.1 “Assessment extensions” in the [Assessment Procedure for Students](#) for eligible grounds for an extension.All requests must be made through the myGriffith portal.
- **Individual Work:** You must complete this assignment individually.
- **Presentation:** You must present/demonstrate your work to a teaching team member for phase 1 in the workshop in week 7 and for phase 2 in the workshop of week 12. Your work will not be marked if you do not present it.
- **Other:** This assignment is a mandatory pass component. You must attain 50% (35/70) (Phases 1 and 2 combined) to be eligible to pass the course.

Introduction

This document will provide an overview of the entire assignment followed by a breakdown of the submission requirements for Phase 1(Due Thursday week 6) and Phase 2 (Due End week 11).

You are required to build a full stack chat application to the specifications listed within this document and from further specifications elicitation from the client.

The chat system will facilitate user's communication with each other, passing messages as text and images in real-time. Some users will have elevated privileges to perform administrative functions such as user creation requests, group and channel assignments.

The development stack to be used is the MEAN stack. Mongo, Express, Angular (20+), Node.js. The real time communication (chat) and other functionality will use socket.io.

The client for this project is the course convenor. In Week 2 you will have an opportunity to ask questions, of the client, to clarify any question you might have about the design and functional requirements for the application. This session will be recorded and provide a source of information for your analysis of the project requirements. The lecture time slot in week 2 will be used for this client briefing. This will be an online meeting.

The full specifications of the app will not be defined in this document. You should read this document in conjunction with the recordings from the client meeting (Week 2). If questions are made after the meeting from any student up until August 5th 5pm (Week4) clarifying requirements they will be published on the course website as an announcement and form part of the project requirements..

Use of AI

As students in a third-year programming course, you are expected to have a solid understanding of fundamental control flow constructs and commonly used data structures across programming languages. In this context, AI tools can be effectively used to support your development process - for example, by assisting with code generation, debugging, or exploring alternative approaches. However, it remains essential that you fully understand any code you submit. You must be able to clearly explain and justify your design and architectural decisions, as well as demonstrate a solid understanding of the syntax and logic involved. Reliance on AI does not replace your responsibility to develop and demonstrate independent programming competence.

Source Version Control

Git must be used during the development of the application. We recommend that you use GitHub and share the repository with your marker. You will be marked on frequent commits to the repository and the usage of git features. All repos should be **kept private** and your teaching staff member added as a collaborator. Your Git repo will show a full history of your progress with the assignment. Working offline for an extended period is not an excuse for limited updates to Git. You must be pushing your code to Git at appropriate intervals during the development period to show appropriate use of the tool.

Phase 1

Phase 1 encompasses the **elicitation of requirements, planning/design** and early **prototype** of the application.

Documentation of your implementation is required. All documentation will use appropriate Markdown syntax and be part of your Git Repository.

For Phase 1 the document will be called "**Phase1.md**".

Your number, name and workshop time should be included at the top of the document.

What documentation is required for submission

1. Brief description of the project (Overview).
2. Outline your strategies for using GIT (branching etc) used during the development of your solution.

3. A clear elicitation of specifications and assumptions used for the application. (functional requirements) (Table layout preferred)
4. Description/listing of data structures used to store and represent the data used in the application.
5. Description of proposed angular Architecture: components, services, models, routes etc (Does not have to be fully implemented but should be defined/thought of.)
6. List of proposed server based endpoints that may be required (not all have to be implemented but should be defined)
7. Design documents (including storyboards using a responsive design methodology)

This documentation lays the groundwork and articulates the design decisions and technical architecture to be used for the full development of the application in Phase 2.

What code is required

The focus for Phase 1 is the UI design of the front end of the application. Backend services will be limited to user management functions.

Provide code implementing a working prototype for the application showing the required UI for each permissions level. (This need not be fully functional but should include a clear layout of the UI elements required as per your design documents). Users, Groups and Channels should be able to be created, assigned and be persistently stored in a JSON file on the server.(Databases will be introduced for Phase 2.) Mock data for other functions can be used. Users should have to do very basic authentication when logging in. (username/password)

Submission

Submit to assessment link on Canvas

1. A single document (PDF) should include your name/snumber and a link to your github repo aswell a copy of the raw markdown document (phase1.md). Make sure your teaching staff member has been added as a collaborator to your Git repo before marking.
2. Any pushes to your repo after the submission date will be ignored.

Marking (Week 7 - workshop)

Marking of this assignment is done via a 1 on 1 interview with your teaching staff member. In the interview you will demonstrate the application and your understanding of the code you have written. You will justify your design decisions.

Your documentation will be reviewed in conjunction with your demonstration.

These sessions will be held in your normal workshop classes (either online or on campus). Failure to attend a marking session will result in your assignment not being marked. Please speak with your teaching staff member **before marking day** if attendance is not possible. An alternate time slot may be arranged within the existing schedule of the subject.

Phase 2

Phase 2 involves the full implementation of the app (frontend, backend, database, sockets) such that it meets all of the required specifications and provides an interface that meets expected UI /UX design guidelines and demonstrates a high level of Usability and accessibility.

Documentation of your implementation is required. All documentation will use appropriate Markdown syntax and be part of your Git Repository.

For Phase 2 the document will be called “**Phase2.md**”

Your name and workshop time should be included at the top of the document.

What documentation is required for submission

1. Clearly documented Specifications/Requirements for the application
2. Full API documentation for server side functionality
3. Full description of Angular components, services and models used
4. Design documents (As per Phase 1, updated if required)
5. Description of testing tools/methodology used and a table listing the automated tests performed.

Submission

Submit to assessment link on Canvas

1. A single document (PDF) should include your name/snumber and a link to your github repo aswell a copy of the raw markdown document (phase2.md). Make sure your teaching staff member has been added as a collaborator to your Git repo before marking.

Marking (Week 12 - workshop)

Marking of this assignment is done via 1 on 1 interview with your teaching staff member. You will demonstrate the application and your understanding of the code. You will justify your design decisions.

Your documentation will be reviewed in conjunction with your demonstration.

These sessions will be held in your normal workshop classes (either online or on campus). Failure to attend a marking session will result in your assignment not being marked. Please speak with your teaching staff member **before marking day** if attendance is not possible. An alternate time slot may be arranged within the existing schedule of the subject.